OL | Organic Letters

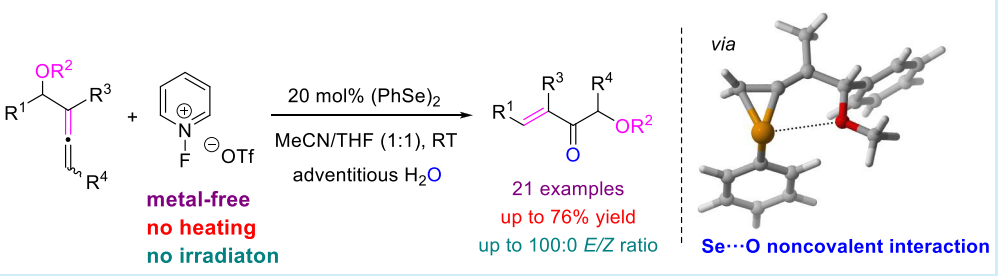
pubs.acs.org/OrgLett Letter

Organoseleno-Catalyzed Synthesis of α,β-Unsaturated α′-Alkoxy Ketones from Allenes Enabled by Se···O Interactions

Mireia Toledano-Pinedo, Teresa Martínez del Campo,* Marta Tiemblo, Israel Fernández,* and Pedro Almendros*

Cite This: *Org. Lett.* 2020, 22, 3979–3984 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information



ABSTRACT: α,β-Unsaturated α′-alkoxy ketones have been prepared under mild conditions from allenes using water as the oxygen source and without the necessity of metals. The organocatalytic oxygenation−rearrangement sequence displays an exquisite chemo-, stereo-, and site-selectivity as well as good functional group compatibility. DFT calculations suggest that stabilizing noncovalent Se···O interactions may be responsible for the observed reactivity.

The α,β-unsaturated α′-alkoxy ketone moiety is a recurrent structural motif in natural products and bioactive molecules which has been also identified as a valuable platform for the preparation of more complex products.[1] Despite the synthesis of the naked α-hydroxy ketone moiety can be smoothly accomplished from alkenes,[2] the preparation of the more functionalized α,β-unsaturated α′-alkoxy ketone scaffold is not an easy task.[3] Consequently, alternative protocols for this framework are highly desirable.

One subject in the continuously growing field of organocatalyzed reactions is the appearance of organoselenium catalysis as a versatile synthetic tool.[4] In particular, electrophilic transformations of isolated alkenes through selenium-based π-acid-type catalysis have been developed (Scheme 1a).[5] Other π-systems have rarely been studied; only Zhao et al. have recently reported the organoseleno-catalyzed functionalization of alkynes (Scheme 1b) and the pyridination of 1,3-dienes (Scheme 1c).[6] Although many efforts have been made in the preparation of seleno derivatives by the stoichiometric reaction of allenes with selenenylating reagents,[7] the use of selenium catalysis in allene chemistry remais unexplored. We wish to report herein the oxygenative rearrangement of the allene skeleton via organoselenium catalysis (Scheme 1d).

During our program focused on studying the reactivity of the allene moiety, an unanticipated observation captured our interest. The reaction of allenyl methyl ether **1a** with stoichiometric amounts of diphenyl diselenide in the presence of 1-fluoropyridinium triflate formed the unexpected α′-alkoxy enone **2a** as the major product (25% yield) along with selenated adducts (Scheme 2a).[7i] Control experiments revealed that both the organoselenium compound and the oxidant were crucial for this transformation. Aimed at exploiting this accidental result in connection with a general synthetic protocol, we speculated that a selenium-catalyzed rearrangement−functionalization sequence of allenyl ethers could be established. For this proposal, we initially decided to reduce the stoichiometric quantity of $(PhSe)_2$ to catalytic amounts. Disappointingly, the use of 10 mol % of $(PhSe)_2$ yielded α′-alkoxy enone **2a** in a diminished 12% yield. The major product (82% yield) of this reaction was enone **3a**, a Meyer−Schuster-type product lacking the methoxy moiety (Scheme 2b). We suspect that the catalytic version using 10 mol % is kinetically slow and higher catalyst loading is required. Noteworthy, when the reaction of allenyl methyl ether **1a** was catalyzed by 20 mol % of $(PhSe)_2$, α,β-unsaturated α′-alkoxy ketone **2a** was obtained in a good 69% yield as a single (*E*)-isomer (Scheme 2c). Similar figures, with just a slight improvement in yield, were obtained using 50 mol % of $(PhSe)_2$ (Scheme 2d). Consequently, we decided that 20 mol % of the organoselenium catalyst should be used as

Received: April 13, 2020
Published: April 27, 2020

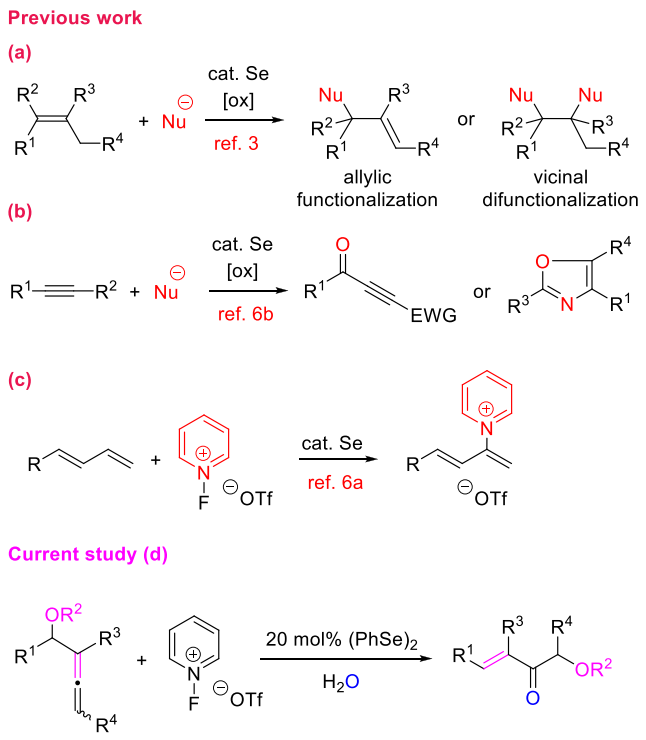


Downloaded via FUDAN UNIV on June 23, 2020 at 14:21:31 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

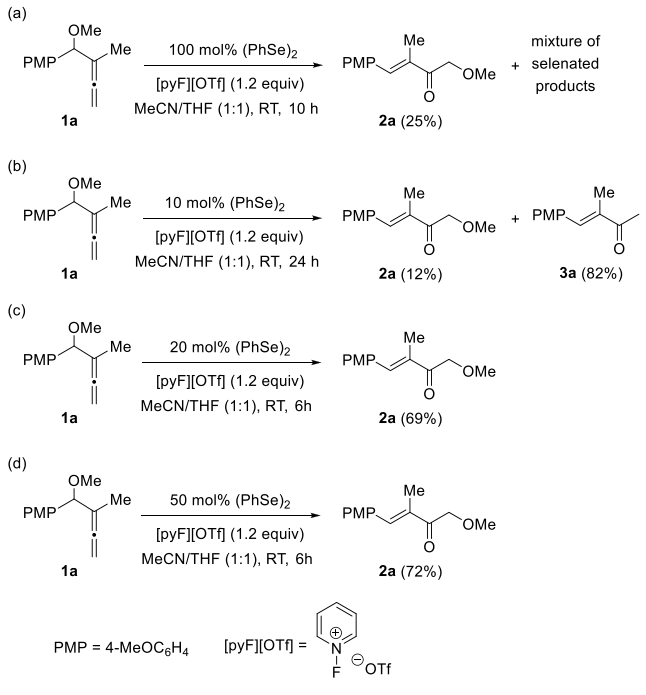
ACS Publications © 2020 American Chemical Society

https://dx.doi.org/10.1021/acs.orglett.0c01288
Org. Lett. 2020, 22, 3979–3984

Scheme 1. State-of-the-Art and Current Work



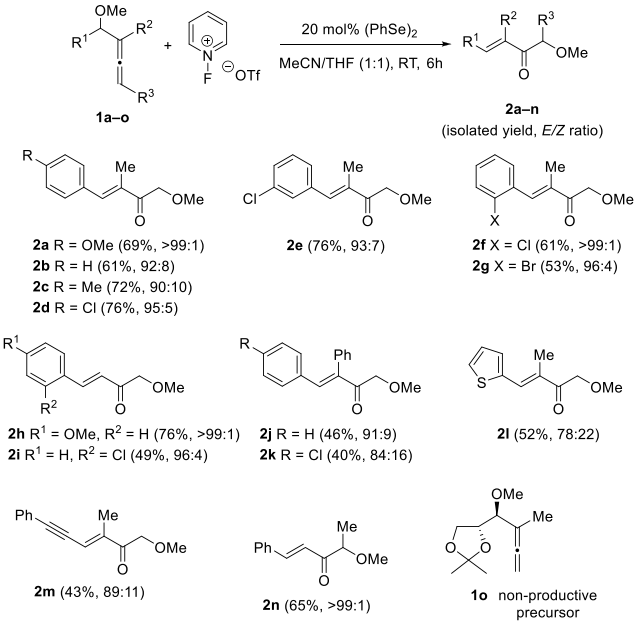
Scheme 2. Impact of the Diphenyl Diselenide Amount on the Selenium-Catalyzed Rearrangement−Functionalization Sequence of Allenyl Methyl Ether 1a



optimized amount for studying the scope of the organoseleno-catalyzed reaction of allenes. When the reaction of precursor **1a** was carried out replacing $(PhSe)_2$ by PhSeCl (20 mol %), a complicated mixture was obtained and product **2a** was obtained in a poor 14% yield.

After fixing a productive catalyst system, different tests were conducted to inspect the substrate scope of the above transformation. The generality of the reaction was first studied using differently substituted allenyl methyl ethers **1**. The corresponding products **2a−m** were achieved in reasonable yields (40−76%) and good or great (*E*)-diastereoselectivity (Scheme 3). The reaction proved to be general because both

Scheme 3. Controlled Synthesis of α,β-Unsaturated α′-Alkoxy Ketones 2a−n



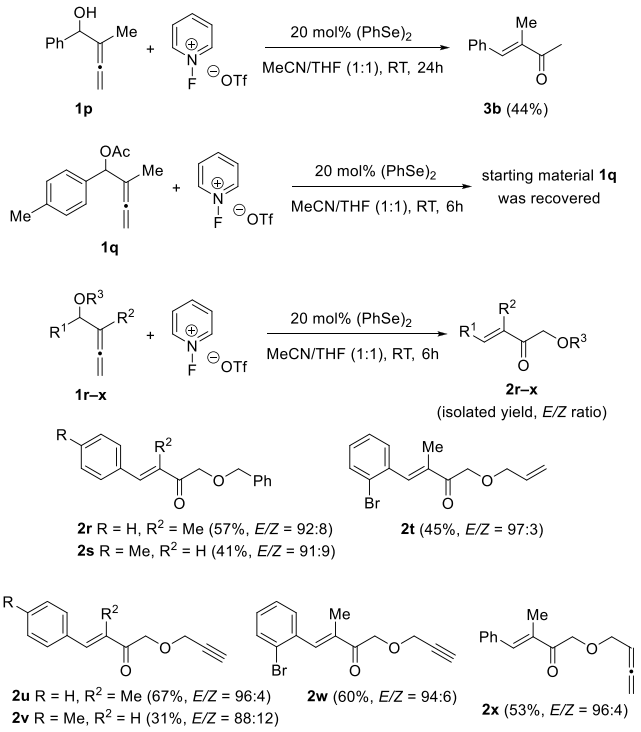
electron-donating (4-MeO and 4-Me) and electron-withdrawing groups (Cl and Br) were well accommodated. Interestingly, the bromine and chlorine moieties on the arene are compatible with enone formation conditions, which may offer further functionalization through selective cross-coupling protocols. Reactions of derivatives **1** bearing substituents on the phenyl nucleus at the *ortho*, *meta*, and *para* positions proceeded smoothly. Hetaryl moieties such as 2-thienyl were also well suited, albeit the *E*/*Z* ratio of **2l** (78:22) was lower compared to other examples. Next, substitution at the allene moiety was examined. Replacement of the methyl group by hydrogen also provided the desired α′-alkoxy enones **2h,i** in fair yields. To judge by the reactions of (1-methoxybuta-2,3-dien-2-yl)benzenes **1j,k**, the steric hindrance imparted by the phenyl group on the cumulene moiety did exert unquestionable influence on enone formation; the reaction of phenyl allenes **1j,k** afforded adducts **2j,k** in a diminished yield (Scheme 3; **2b,d** vs **2j,k**). Noteworthy, the alkyne portion may also be tolerated and methyl ether **1m** provided α′-methoxy conjugated enynone **2m** with exquisite chemoselectivity (allene vs alkyne). Noteworthy, when allenyl methyl ether **1n** decorated with a methyl substituent at the terminal allene carbon was used as the starting material, the reaction also worked well to provide the α′-methyl α′-methoxy *gem*-substituted enone **2n** (Scheme 3). However, when the protocol was applied to alkyl-substituted allenyl methyl ether **1o**, enone formation could not be achieved (Scheme 3). The (*E*)-stereochemistry of enones **2** was established by means of NOE experiments performed in compound **2c** (see the Supporting Information).

Next, we sought to develop a more general protocol by replacing the methyl ether framework by different oxygenated

https://dx.doi.org/10.1021/acs.orglett.0c01288
Org. Lett. 2020, 22, 3979−3984

functionalities. Initial efforts were a little bit frustrating because the reaction of allenol **1p** yielded enone **3b**, while allenyl acetate **1q** was unreactive.[9] Fortunately, allenyl ethers **1r–x** bearing extra benzyl, allyl, propargyl, and cumulenyl moieties provided the desired rearranged functionalized adducts **2r–x** in synthetically useful yields (Scheme 4). It should be noted

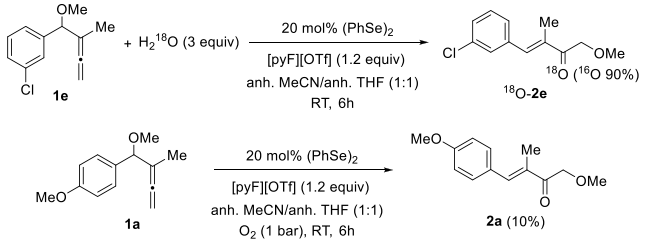
Scheme 4. Controlled Synthesis of α,β-Unsaturated α′-Alkoxy Ketones 2r–x



that the mild reaction conditions allow for a complete chemoselectivity, with the unsaturated allyloxy, propargyloxy, and allenyloxy components present in allene precursors **1** being transferred unaltered to the final rearranged products **2**. Simultaneously, functionalized α′-alkoxy enones (*E*)-**2r–x** were stereoselectively formed in a high *E*/*Z* ratio (up to 97:3) comparable to methyl ethers **2a–n**.

The accessible water of regular (HPLC grade) acetonitrile and THF was adequate for the reaction to progress profitably. In order to confirm the participation of adventitious water in the formation of polyfunctional adducts **2**, an ^{18}O-labeling experiment was conducted (Scheme 5). Indeed, the use of $H_2{}^{18}O$ in the reaction of **1e** in anhydrous solvents resulted in the formation of ^{18}O-labeled **2e**, as observed by high-resolution mass spectrometry (see the Supporting Information). Besides,

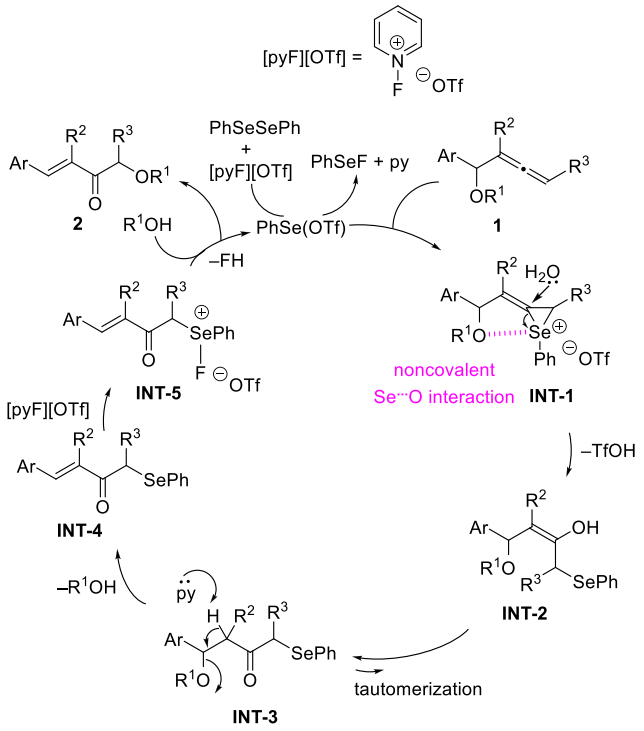
Scheme 5. Labeling and Control Studies



the possible participation of atmospheric oxygen in the formation of the carbonyl group can be ruled out in view of the observed drastic reduction of the yield of enone **2a** when conducting a control reaction in dehydrated solvents in the presence of an O_2 balloon (Scheme 5).

A possible pathway for the generation of functionalized enones **2** from allenes **1** through organoselenium catalysis is outlined in Scheme 6. First, and similar to the pathway

Scheme 6. Rationalization for the Formation of α,β-Unsaturated α′-Alkoxy Ketones 2 by Organoselenium Catalysis



proposed by Zhao and co-workers,[6a] the selenium-based π-acid species PhSe(OTf) is originated from PhSeSePh by way of the oxidative reagent [pyF][OTf]. Subsequently, an alkenyl seleniranium **INT-1** may be formed through regioselective carbophilic interaction to the terminal allene double bond. The proposed formation of the cyclic selenium intermediate **INT-1** in a *syn* manner toward the OR^1 group is in agreement with recent reports on stabilizing noncovalent Se···O interactions.[8,9] Then, controlled water attack to the three-membered heterocyclic species with concomitant release of ring strain and triflic acid liberation should afford enol **INT-2**, which is in equilibrium with its more stable keto-tautomer **INT-3**. The presence of free pyridine in the reaction medium should provoke the abstraction of the acidic hydrogen, that is associated with alcohol elimination and formation of selenoenone **INT-4**. Further [pyF][OTf]-mediated oxidation of the PhSe group in **INT-4** should give rise to selenonium species **INT-5**.[10] α,β-Unsaturated α′-alkoxy ketones **2** would be finally formed by S_N2 displacement of the phenylselenonium moiety by the in situ generated alcohol, with concurrent loss of FH and regeneration of the selenium–π-acid catalyst.

Density functional theory (DFT) calculations at the dispersion-corrected PCM(MeCN)-B3LYP-D3/def2-

https://dx.doi.org/10.1021/acs.orglett.0c01288
Org. Lett. 2020, 22, 3979–3984

TZVPP//PCM(MeCN)-B3LYP-D3/def2-SVP level[11] were carried out to gain more insight into the suggested occurrence of a Se···O noncovalent interaction in the initial stages of the transformation. To this end, we computed the reaction involving allene **1b** and PhSeOTf which leads to the corresponding seleniranium **INT-1b** (see Figure 1). Our

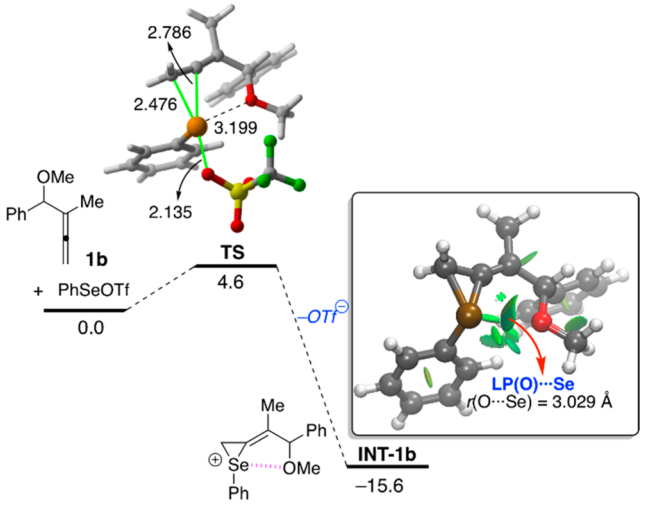


Figure 1. Computed reaction profile for the formation of intermediate **INT-1b** from **1b** and PhSeOTf. Relative free energies and bond lengths are given in kcal/mol and angstroms, respectively. All data have been computed at the PCM(MeCN)-B3LYP-D3/def2-TZVPP//PCM(MeCN)-B3LYP-D3/def2-SVP level. Inset: Contour plots of the reduced density gradient isosurfaces (density cutoff of 0.03 au) for compound **INT-1b**. The green surfaces indicate attractive noncovalent interactions.

calculations indicate that the formation of **INT-1b** occurs concertedly via **TS**, a saddle point associated with the simultaneous (yet asynchronous) formation of the new C–Se bonds and the rupture of the Se–OTf bond. The low barrier ($\Delta G^{\ddagger}$ = 4.6 kcal/mol) and the high exergonicity (ΔG_R = −15.6 kcal/mol) computed for this reaction is fully compatible with a process occurring at room temperature. Interestingly, the Se···O distance in either **TS** or intermediate **INT-1b** (3.199 and 3.029 Å, respectively) are shorter than the sum of their respective van der Waals radii (3.42 Å), therefore suggesting a significant noncovalent interaction involving these atoms. The occurrence of this stabilizing intramolecular interaction can be computationally visualized and confirmed by means of the NCIPLOT method.[12] As clearly shown in the inset of Figure 1, there exists a significant noncovalent attractive interaction (green surface) between the oxygen and selenium atoms in **INT-1b**.[13]

Moreover, the crucial role of this LP(O)···Se noncovalent interaction in the process is supported by comparing the reaction profile involving the OAc counterpart of **1b** (see Figure S1). For this system, the Se···O distance of the corresponding transition state is much longer (3.333 Å) therefore indicating a weaker Se···O interaction as a consequence of the reduced donor ability of the oxygen atom. As a result, the computed barrier for this transformation is significantly higher ($\Delta G^{\ddagger}$ = 7.4 kcal/mol) and the process becomes much less exergonic (ΔG_R = −8.5 kcal/mol) than the reaction involving its OMe counterpart **1b**.

To summarize, an organocatalyzed direct synthesis of α′-alkoxy enones from allenyl ethers under mild conditions with adventitious water as the oxygen source and without the involvement of metals has been satisfactorily accomplished. This chemo-, stereo-, and site-selective oxygenative rearrangement process involves an organoselenium-based π-acid-type catalysis. DFT calculations provide further support to the occurrence of a stabilizing LP(O)···Se noncovalent interaction which significantly facilitates the transformation.[14]

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.0c01288.

Experimental procedures, characterization data of new compounds, copies of NMR spectra, computational details, and alternative reaction pathway (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Teresa Martínez del Campo − *Grupo de Lactamas y Heterociclos Bioactivos, Departamento de Química Orgánica, Unidad Asociada al CSIC, Facultad de Química, Universidad Complutense de Madrid, 28040 Madrid, Spain*; Email: tmcampo@quim.ucm.es

Israel Fernández − *Departamento de Química Orgánica and Centro de Innovación en Química Avanzada (ORFEO–CINQA), Facultad de Química, Universidad Complutense de Madrid, 28040 Madrid, Spain*; orcid.org/0000-0002-0186-9774; Email: israel@quim.ucm.es

Pedro Almendros − *Instituto de Química Orgánica General, IQOG, CSIC, 28006 Madrid, Spain*; orcid.org/0000-0001-6564-2758; Email: palmendros@iqog.csic.es

Authors

Mireia Toledano-Pinedo − *Grupo de Lactamas y Heterociclos Bioactivos, Departamento de Química Orgánica, Unidad Asociada al CSIC, Facultad de Química, Universidad Complutense de Madrid, 28040 Madrid, Spain*

Marta Tiemblo − *Grupo de Lactamas y Heterociclos Bioactivos, Departamento de Química Orgánica, Unidad Asociada al CSIC, Facultad de Química, Universidad Complutense de Madrid, 28040 Madrid, Spain*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.orglett.0c01288

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported in part by the Spanish Agencia Estatal de Investigación and European Regional Development Fund (Projects PGC2018-095025-B-I00, CTQ2016-78205-P, and CTQ2016-81797-REDC). We are grateful to Prof. B. Alcaide for continued support.

■ DEDICATION

Dedicated to Professor Benito Alcaide on the occasion of his 70th birthday

■ REFERENCES

(1) For occurrence in natural products, see: (a) Stierle, D. B.; Stierle, A. A.; Ganser, B. New Phomopsolides from a *Penicillium* sp. *J. Nat.*

Prod. **1997**, *60*, 1207−1209. (b) Clericuzio, M.; Mella, M.; Vita-Finzi, P.; Zema, M.; Vidari, G. Cucurbitane Triterpenoids from *Leucopaxillus gentianeus*. *J. Nat. Prod.* **2004**, *67*, 1823−1828. (c) Jang, K. H.; Lee, B. H.; Choi, B. W.; Lee, H. S.; Shin, J. Chromenes from the Brown Alga *Sargassum siliquastrum*. *J. Nat. Prod.* **2005**, *68*, 716−723 For synthetic utility, see:. (d) Fujita, M.; Hiyama, T. Fluoride Ion Catalyzed Reduction of Aldehydes and Ketones with Hydrosilanes. Synthetic and Mechanistic Aspects and an Application to the Threo-Directed Reduction of α-Substituted Alkanones. *J. Org. Chem.* **1988**, *53*, 5405−5415. (e) Mulzer, J.; List, B.; Bats, J. W. Stereocontrolled Synthesis of a Nonracemic Vitamin B12 A-B-Semicorrin. *J. Am. Chem. Soc.* **1997**, *119*, 5512−5518. (f) Harada, S.; Taguchi, T.; Tabuchi, N.; Narita, K.; Hanzawa, Y. Acylzirconocene Chloride as an Unmasked Acyl Anion. *Angew. Chem., Int. Ed.* **1998**, *37*, 1696−1698. (g) Palomo, C.; Oiarbide, M.; García, J. M.; González, A.; Lecumberri, A.; Linden, A. A Chiral Acrylate Equivalent for Metal-Free Diels-Alder Reactions: endo-2-Acryloylisoborneol. *J. Am. Chem. Soc.* **2002**, *124*, 10288−10289. (h) Rodriquez, M.; Terracciano, S.; Cini, E.; Settembrini, G.; Bruno, I.; Bifulco, G.; Taddei, M.; Gomez-Paloma, L. Total Synthesis, NMR Solution Structure, and Binding Model of the Potent Histone Deacetylase Inhibitor FR235222. *Angew. Chem., Int. Ed.* **2006**, *45*, 423−427. (i) Ichikawa, Y.; Egawa, H.; Ito, T.; Isobe, M.; Nakano, K.; Kotsuki, H. Stereocontrolled Route to Vicinal Diamines by [3.3] Sigmatropic Rearrangement of Allyl Cyanate: Asymmetric Synthesis of *anti*-(2*R*,3*R*)- and *syn*-(2*R*,3*S*)-2,3-DiaminobutanoicAcids. *Org. Lett.* **2006**, *8*, 5737−5740. (j) Palomo, C.; Oiarbide, M.; García, J. M. α-Hydroxy Ketones as Useful Templates in Asymmetric Reactions. *Chem. Soc. Rev.* **2012**, *41*, 4150−4164.

(2) (a) Plietker, B. Alkenes as Ketol Surrogates−A NewApproach toward Enantiopure Acyloins. *Org. Lett.* **2004**, *6*, 289−291. (b) Plietker, B. The RuO_4-Catalyzed Ketohydroxylation. Part 1. Development, Scope, and Limitation. *J. Org. Chem.* **2004**, *69*, 8287−8296. (c) Plietker, B. The RuO_4-Catalyzed Ketohydroxylation, Part II: A Regio-, Chemo- and Stereoselectivity Study. *Eur. J. Org. Chem.* **2005**, *2005*, 1919−1929.

(3) (a) Pierre, F.; Enders, D. Nucleophilic Alkynoylation of Aldehydes with Metalated α-Aminonitriles: Regioselective Synthesis of α-Hydroxyenones. *Tetrahedron Lett.* **1999**, *40*, 5301−5305. (b) Ishikawa, T.; Aikawa, T.; Ohata, E.; Iseki, T.; Maeda, S.; Matsuo, T.; Fujino, T.; Saito, S. Two-Directional Elaboration of Hydroxyacetone under Thermodynamically Controlled Conditions: Allylation or 2-Propynylation and Aldol Reaction. *J. Org. Chem.* **2007**, *72*, 435−441. (c) Christoffers, J.; Werner, T.; Frey, W.; Baro, A. Straightforward Synthesis of (*R*)-(−)-Kjellmanianone. *Chem. - Eur. J.* **2004**, *10*, 1042−1045. (d) Cosp, A.; Dresen, C.; Pohl, M.; Walter, L.; Röhr, C.; Müller, M. α. β-Unsaturated Aldehydes as Substrates for Asymmetric C−C Bond Forming Reactions with Thiamin Diphosphate (ThDP)-Dependent Enzymes. *Adv. Synth. Catal.* **2008**, *350*, 759−771.

(4) For reviews, see: (a) Shao, L.; Li, Y.; Lu, J.; Jiang, X. Recent Progress in Selenium-Catalyzed Organic Reactions. *Org. Chem. Front.* **2019**, *6*, 2999−3041. (b) Guo, R.; Liao, L.; Zhao, X. Electrophilic Selenium Catalysis with Electrophilic N-F Reagents as the Oxidants. *Molecules* **2017**, *22*, 835 DOI: 10.3390/molecules22050835.

(5) For a review, see: (a) Ortgies, S.; Breder, A. Oxidative Alkene Functionalizations via Selenium-π-Acid Catalysis. *ACS Catal.* **2017**, *7*, 5828−5840 For selected recent references, see:. (b) Cresswell, A.; Eey, J. S. T.-C.; Denmark, S. E. Catalytic, Stereospecific *syn*-Dichlorination of Alkenes. *Nat. Chem.* **2015**, *7*, 146−152. (c) Kawamata, Y.; Hashimoto, T.; Maruoka, K. A Chiral Electrophilic Selenium Catalyst for Highly Enantioselective Oxidative Cyclization. *J. Am. Chem. Soc.* **2016**, *138*, 5206−5209. (d) Guo, R.; Huang, J.; Zhao, X. Organoselenium-Catalyzed Oxidative Allylic Fluorination with Electrophilic N−F Reagent. *ACS Catal.* **2018**, *8*, 926−930. (e) Rode, K.; Palomba, M.; Ortgies, S.; Rieger, R.; Breder, A. Aerobic Allylation of Alcohols with Non-Activated Alkenes Enabled by Light-Driven Selenium-π-Acid Catalysis. *Synthesis* **2018**, *50*, 3875−3885. (f) Gilbert, B. B.; Eey, S. T.-C.; Ryabchuk, P.; Garry, O.; Denmark, S. E. Organoselenium-Catalyzed Enantioselective *syn*-Dichlorination of Unbiased Alkenes. *Tetrahedron* **2019**, *75*, 4086−4098. (g) Otsuka, Y.; Shimazaki, Y.; Nagaoka, H.; Maruoka, K.; Hashimoto, T. Scalable Synthesis of a Chiral Selenium π-Acid Catalyst and Its Use in Enantioselective Iminolactonization of β,γ-Unsaturated Amides. *Synlett* **2019**, *30*, 1679−1682. (h) Tao, Z.; Gilbert, B. B.; Denmark, S. E. Catalytic, Enantioselective *syn*-Diamination of Alkenes. *J. Am. Chem. Soc.* **2019**, *141*, 19161−19170. (i) Krätzschmar, F.; Ortgies, S.; Willing, Y. N.; Breder, A. Rational Design of Chiral Selenium-π-Acid Catalysts. *Catalysts* **2019**, *9*, 153. (j) Tabor, R.; Obenschain, D. C.; Michael, F. E. Selenophosphoramide-Catalyzed Diamination and Oxyamination of Alkenes. *Chem. Sci.* **2020**, *11*, 1677−1682. (k) Wang, X.; Wang, Q.; Xue, Y.; Sun, K.; Wub, L.; Zhang, B. An organoselenium-Catalyzed N1 - and N2 -Selective Aza-Wacker Reaction of Alkenes with Benzotriazoles. *Chem. Commun.* **2020**, *56*, 4436.

(6) (a) Liao, L.; Guo, R.; Zhao, X. Organoselenium-Catalyzed Regioselective C−H Pyridination of 1,3-Dienes and Alkenes. *Angew. Chem., Int. Ed.* **2017**, *56*, 3201−3205. (b) Liao, L.; Guo, R.; Zhao, X. Selenium-π-Acid Catalyzed Oxidative Functionalization of Alkynes: Facile Access to Ynones and Multisubstituted Oxazoles. *ACS Catal.* **2018**, *8*, 6745−6750.

(7) For selected references, see: (a) Ogawa, A.; Obayashi, R.; Doi, M.; Sonoda, N.; Hirao, T. A Novel Photoinduced Thioselenation of Allenes by Use of a Disulfide−Diselenide Binary System. *J. Org. Chem.* **1998**, *63*, 4277−4281. (b) Chen, G.; Fu, C.; Ma, S. Studies on Electrophilic Addition Reaction of 2,3-Allenoates with PhSeCl. *Tetrahedron* **2006**, *62*, 4444−4452. (c) Chen, G.; Fu, C.; Ma, S. Electrophilic Interaction of 2,3-Allenoates with PhSeCl. An Unexpected Highly Stereoselective Synthesis of 3-Phenylseleno-4-oxo-2(*E*)-alkenoates. *J. Org. Chem.* **2006**, *71*, 9877−9879. (d) He, G.; Guo, H.; Qian, R.; Guo, Y.; Fu, C.; Ma, S. Studies on Highly Regio- and Stereoselective Selenohydroxylation Reaction of 1,2-Allenyl Phosphine Oxides with PhSeCl. *Tetrahedron* **2009**, *65*, 4877−4889. (e) Ma, S. Control of Regio- and Stereoselectivity in Electrophilic Addition Reactions of Allenes. *Pure Appl. Chem.* **2007**, *79*, 261−267. (f) He, G.; Yu, Y.; Fu, C.; Ma, S. Highly Selective Synthesis of [(Z)-3-Chloro-2-(phenylselanyl)-1-alkenyl]phosphonates and 2-Ethoxy-4-(phenylselanyl)-2,5-dihydro-1,2-oxaphosphole 2-Oxides by Electrophilic Reaction of 1,2-Alkadienylphosphonates with PhSeCl. *Eur. J. Org. Chem.* **2010**, *2010*, 101−110. (g) Alcaide, B.; Almendros, P.; Luna, A.; Gómez-Campillos, G.; Torres, M. R. Ring Enlargement versus Selenoetherification on the Reaction of Allenyl Oxindoles with Selenenylating Reagents. *J. Org. Chem.* **2012**, *77*, 3549−3556. (h) Wang, J.; Wei, C.; Li, X.; Zhao, P.; Shan, C.; Wojtas, L.; Chen, H.; Shi, X. Gold Redox Catalysis with a Selenium Cation as Mild Oxidant. *Chem. - Eur. J.* **2020**, DOI: 10.1002/chem.202000166. (i) Alcaide, B.; Almendros, P.; Martínez del Campo, T.; Martín, L.; Palop, G.; Toledano-Pinedo, M. Oxidative Selenofunctionalization of Allenes: Convenient Access to 2-(Phenylselanyl)-but-2-enals and 4-Oxo-3-(phenylselanyl)pent-2-enoates. *Org. Chem. Front.* **2019**, *6*, 2447−2451.

(8) (a) Wang, W.; Zhu, H.; Liu, S.; Zhao, Z.; Zhang, L.; Hao, J.; Wang, Y. Chalcogen−Chalcogen Bonding Catalysis Enables Assembly of Discrete Molecules. *J. Am. Chem. Soc.* **2019**, *141*, 9175−9179. (b) Ungati; Govindaraj, V.; Narayanan, M.; Mugesh, G. Probing the Formation of a Seleninic Acid in Living Cells by the Fluorescence Switching of a Glutathione Peroxidase Mimetic. *Angew. Chem., Int. Ed.* **2019**, *58*, 8156−8160. (c) Wonner, P.; Steinke, T.; Vogel, L.; Huber, S. M. Carbonyl Activation by Selenium- and Tellurium-Based Chalcogen Bonding in a Michael Addition Reaction. *Chem. - Eur. J.* **2020**, *26*, 1258−1262. (d) Wang, W.; Zhu, H.; Feng, L.; Yu, Q.; Hao, J.; Zhu, R.; Wang, Y. Dual Chalcogen−Chalcogen Bonding Catalysis. *J. Am. Chem. Soc.* **2020**, *142*, 3117−3124. (e) Young, H. C. M.; Elmi, A.; Pascoe, D. J.; Morris, R. K.; McLaughlin, C.; Woods, A. M.; Frost, A. B.; de la Houpliere, A.; Ling, K. B.; Smith, T. K.; Slawin, A. M. Z.; Willoughby, P. H.; Cockroftb, S. L.; Smith, A. D. The Importance of 1,5-Oxygen···Chalcogen Interactions in Enantioselective Isochalcogenourea Catalysis. *Angew. Chem., Int. Ed.* **2020**, *59*, 3705−3710.

https://dx.doi.org/10.1021/acs.orglett.0c01288
Org. Lett. 2020, 22, 3979−3984

(9) The failure of allene **1q**, bearing an AcO group instead of an alkoxy moiety, may be tentatively explained through a weaker Se···O interaction, which should not impart stabilization enough to the putative phenylseleniranium intermediate **INT-1q**.
(10) The application of selenonium cations as Lewis acids in organocatalytic reactions has been recently documented: He, X.; Wang, X.; Tse, Y.-L.; Ke, Z.; Yeung, Y.-Y. Applications of Selenonium Cations as Lewis Acids in Organocatalytic Reactions. *Angew. Chem., Int. Ed.* **2018**, *57*, 12869−12873.
(11) See the Computational Details in the Supporting Information.
(12) Johnson, E. R.; Keinan, S.; Mori-Sánchez, P.; Contreras-García, J.; Cohen, A. J.; Yang, W. Revealing Noncovalent Interactions. *J. Am. Chem. Soc.* **2010**, *132*, 6498−6506.
(13) A similar interaction is found by the NCIPLOT method for the saddle point **TS**.
(14) Taking into account both that electrophiles are usually added to the internal double bond of the allene moiety[15] as well as the possibility of isomerization of 2,3-allenols to 2-oxy-1,3-diene under certain conditions,[16] another possible mechanism (see Scheme S1 in the Supporting Information) may be contemplated: Precursors **1** should be converted under acidic conditions to allene-type allyl cations **I** through OP release. Next, π-bond migration followed by water attack should form 2-oxy-1,3-dienes **II**, which can isomerize to enones **3**. Next, $PhSe^+$ can react with intermediates **II** to form α-PhSe-ketones **INT-4**, which followed by oxidation and displacement should generate α,β-unsaturated α'-alkoxy ketones **2**. The formation of compounds **3a,b** might go through the picture depicted in Scheme S1. However, we think that this alternative path is not operative for the formation of compounds **2a−n** and **2r−x**, because it may not account for the different reactivity observed for AcO-allene **1q** and alkoxyallenes **1a−n** and **1r−x**.
(15) (a) Fu, C.; Li, J.; Ma, S. Efficient Two-Step Synthesis of 3-Halo-3-enals or 2-Halo-2-alkenyl ketones from Propargylic Bromides *via* a Unique Cationic 1,2-Aryl or Proton Shift in Electrophilic Addition Reaction of 2,3-Allenols with X^+. *Chem. Commun.* **2005**, 4119−4121. (b) Alcaide, B.; Almendros, P.; Luna, A.; Cembellín, S.; Arnó, M.; Domingo, L. R. Controlled Rearrangement of Lactam-Tethered Allenols with Brominating Reagents. A Combined Experimental and Theoretical Study on the α-Keto- *versus* β-Keto Lactam Formation. *Chem. - Eur. J.* **2011**, *17*, 11559−11566. (c) Murai, K.; Shimizu, N.; Fujioka, H. Enantioselective Iodolactonization of Allenoic Acids. *Chem. Commun.* **2014**, *50*, 12530−12533.
(16) (a) Zhang, X.; Fu, C.; Ma, S. Highly Stereoselective Facile Synthesis of 2-Acetoxy-1,3(*E*)-alkadienes via a Rh(I)-Catalyzed Isomerization of 2,3-Allenyl Carboxylates. *Org. Lett.* **2011**, *13*, 1920−1923. (b) Jin, Z.; Hidinger, R. S.; Xu, B.; Hammond, G. B. Stereoselective Synthesis of Monofluoroalkyl α,β-Unsaturated Ketones From Allenyl Carbinol Esters Mediated by Gold and Selectfluor. *J. Org. Chem.* **2012**, *77*, 7725−7729.

https://dx.doi.org/10.1021/acs.orglett.0c01288
Org. Lett. 2020, 22, 3979−3984